

DOCUMENTATION

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QUESTION NO -2

. Introduction

A photography company wants to build a simple image editing system that provides essential photo editing features similar to mobile apps. This project demonstrates how **OpenCV** and **NumPy** can be used to implement brightness adjustment, noise removal, edge detection, feature detection, and panorama stitching.

Objectives

The system should:

1. Change image brightness using thresholding
2. Remove noise using filters
3. Detect edges in images
4. Find important points/features in images
5. Join multiple images together to create a panorama

Tools & Libraries

- **Python** → Programming language
- **OpenCV (cv2)** → Image processing library
- **NumPy** → Numerical operations on image arrays
- **Matplotlib** → Display images in Jupyter Notebook

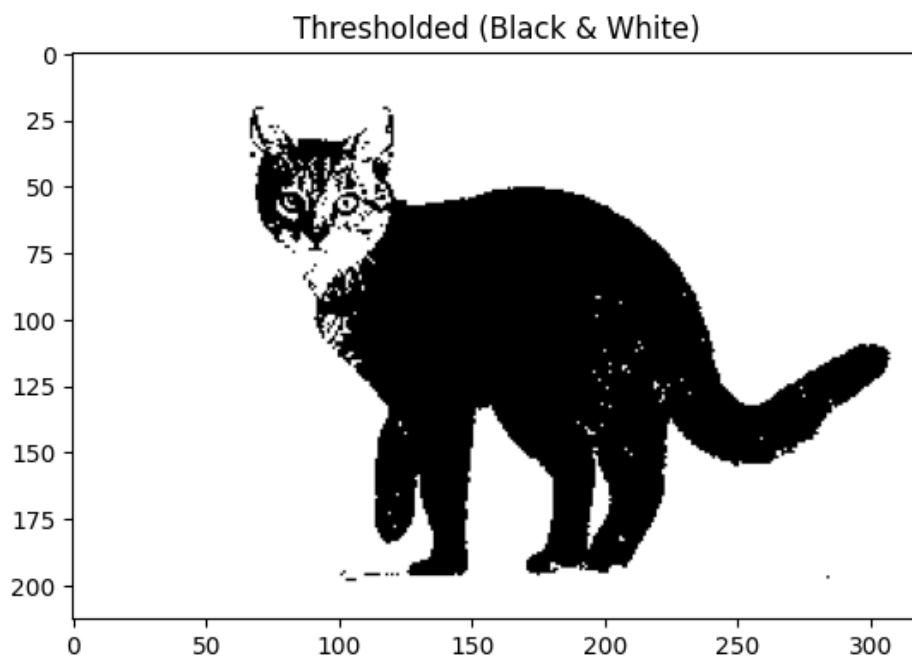
Implementation with screenhote

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# 1. Change Image Brightness using Thresholding
img = cv2.imread("C:\\Users\\nstip\\Downloads\\sample.jpg.jpeg", cv2.IMREAD_GRAYSCALE)
ret, binary = cv2.threshold(img, 127, 255, cv2.THRESH_BINARY)

plt.imshow(binary, cmap="gray")
plt.title("Thresholded (Black & White)")
plt.axis("on")
plt.show()
```



```
blurred = cv2.medianBlur(img, 5) # Median filter removes salt-and-pepper noise  
plt.imshow(blurred, cmap="gray")  
plt.title("Noise Removed")  
plt.axis("off")  
plt.show()
```

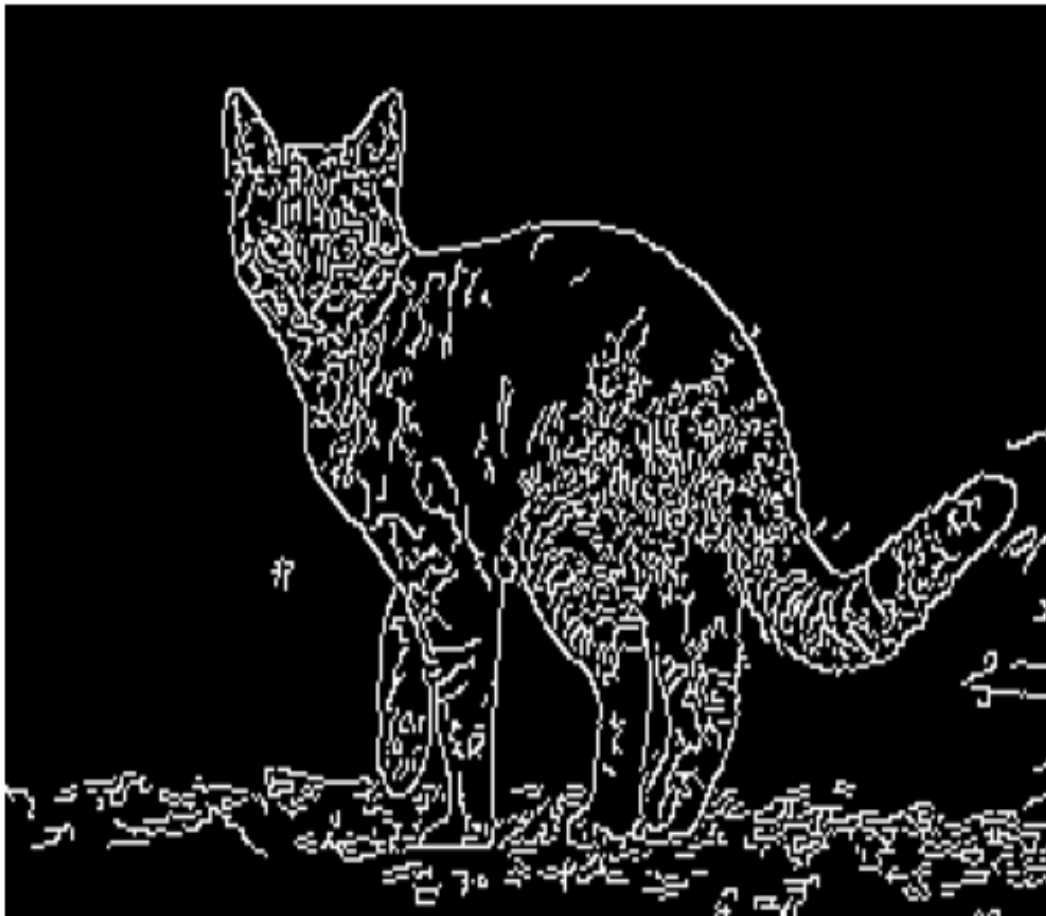
Noise Removed



```
edges = cv2.Canny(img, 100, 200)  
plt.imshow(edges, cmap="gray")  
plt.title("Edges Detected")  
plt.axis("off")  
plt.show()
```

```
edges = cv2.Canny(img, 100, 200)
plt.imshow(edges, cmap="gray")
plt.title("Edges Detected")
plt.axis("off")
plt.show()
```

Edges Detected



```
orb = cv2.ORB_create()
keypoints, descriptors = orb.detectAndCompute(img, None)
img_features = cv2.drawKeypoints(img, keypoints, None, color=(0,205,0,204))

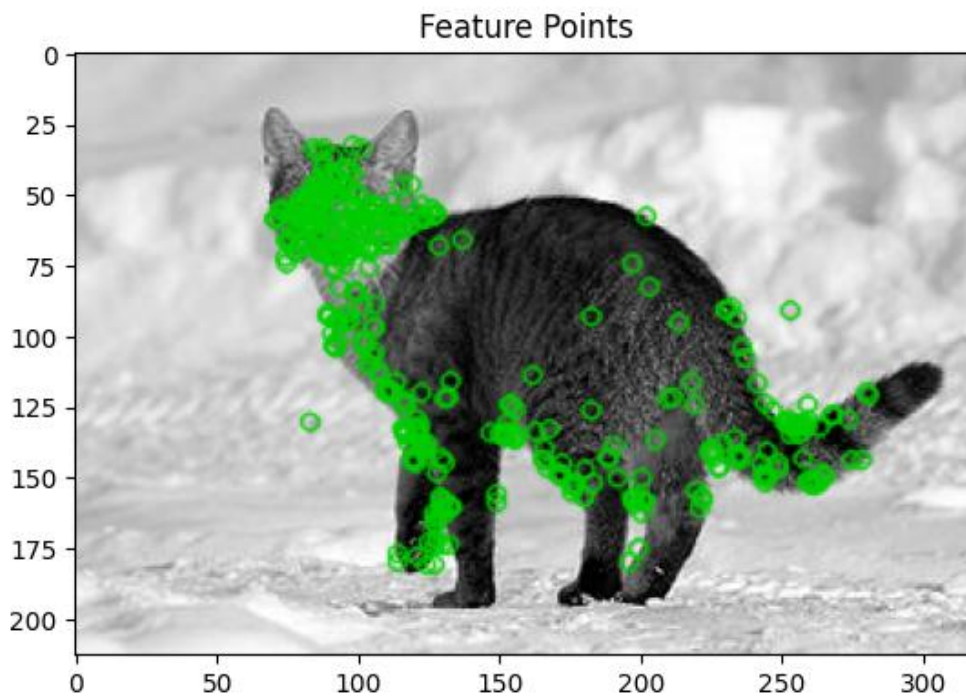
plt.imshow(img_features)
plt.title("Feature Points")
plt.axis("on")
plt.show()
```

```

: orb = cv2.ORB_create()
keypoints, descriptors = orb.detectAndCompute(img, None)
img_features = cv2.drawKeypoints(img, keypoints, None, color=(0,205,0,204))

plt.imshow(img_features)
plt.title("Feature Points")
plt.axis("on")
plt.show()

```



```

: img1 = cv2.imread("left.jpg")
img2 = cv2.imread("right.jpg")

```

write: One or both images not found or path incorrect

```

: img1 = cv2.imread("C:\\Users\\nstip\\Downloads\\left.jpg")
img2 = cv2.imread("C:\\Users\\nstip\\Downloads\\right.jpg")

```

```
] : if img1 is None or img2 is None:  
    print("write: One or both images not found or path incorrect")  
else:  
    print("Images loaded successfully")
```

write: One or both images not found or path incorrect

```
] : print("Image1 shape:", None if img1 is None else img1.shape)  
    print("Image2 shape:", None if img2 is None else img2.shape)
```

Image1 shape: None

Image2 shape: None

```
] :
```

THANK YOU